

Generative AI as Seniority-Biased Technological Change: Evidence from U.S. Résumé and Job Posting Data*

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Abstract

This paper studies whether generative AI (GenAI) constitutes *seniority-biased technological change*, disproportionately reducing demand for junior workers. We develop a conceptual framework in which GenAI adoption reduces junior labor demand through task displacement and labor-saving productivity gains. We test the framework’s mechanisms and implications using U.S. résumé data covering 65 million workers at more than 280,000 firms, allowing us to track firm-level employment by seniority. GenAI adoption is identified through text analysis that detects ‘GenAI integrator’ job postings, signaling active GenAI implementation by firms. Following adoption, junior employment declines in adopting firms relative to non-adopters, while senior employment trends remain largely unchanged. This decline is concentrated in occupations most exposed to GenAI and is driven primarily by slower hiring rather than increased separations. Moreover, in adopting firms, GenAI-exposed tasks become increasingly less likely to appear in junior task bundles.

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tions and zero for low-exposure occupations; γ_{it} are firm-by-time fixed effects; and $\xi_{p(i)ct}$ are industry-by-exposure-by-time fixed effects. Equation 12 has the same structure as the seniority-based triple-difference in Equation 11, but here the third dimension is *occupational exposure* (high vs. low) rather than seniority. The coefficients β_j^s trace, separately for juniors and seniors, how employment in high-exposure occupations evolves relative to low-exposure occupations within the same firm and quarter, comparing adopters to non-adopters.

Figure 6 plots the estimated β_j^s coefficients for juniors and seniors, with both series normalized to zero in 2022Q4, the last pre-ChatGPT quarter. Two patterns stand out. First, throughout the pre-ChatGPT period, high-exposure occupations expand faster than low-exposure occupations at adopters relative to non-adopters for both seniority groups, consistent with adopters tilting their workforce toward exposed occupations over time. The two profiles track each other closely through 2022, with the senior coefficients rising somewhat faster than the junior coefficients. Second, beginning in 2023Q1, the two series diverge sharply. The junior series declines steadily, reaching a roughly 7 log point decline in high-exposure versus low-exposure junior employment at adopters by 2025Q1, relative to the 2022Q4 baseline. In contrast, the senior series continues to rise after 2022Q4, with no visible break around the ChatGPT launch.

Taken together, these results provide two insights. First, the post-2022 contraction in junior employment at adopters is concentrated in high-exposure occupations rather than reflecting a broad-based decline across all junior roles. Second, the design helps rule out a purely compositional explanation—namely, that high-exposure occupations declined economy-wide after 2023 and adopters simply employed a larger share of such roles. By comparing high- and low-exposure occupations *within* the same firm-by-quarter, and by running the same specification for seniors as a placebo-like comparison, we find that high-exposure occupations contract relative to low-exposure occupations only for juniors and only at adopters after the ChatGPT launch. The senior profile continues along its pre-existing upward trend, consistent with the interpretation that GenAI adoption is seniority-biased: it shifts demand away from junior workers in the very occupations where the technology is most applicable, while leaving senior employment in those same occupations essentially unaffected.

Supplemental Appendix A.3 reports a complementary heterogeneity exercise that re-